

Use Case 1: Manage User Session

Use Case Goal: To provide secure, role-based access to the system.

Actors: Admin / Owner, Front Desk Staff

Preconditions: The user has an active account with an assigned role and branch.

Trigger: The user accesses the system login page.

Included Use Case: Open Role Dashboard

Extension Use Cases: Log Out; Manage Own Profile

Main Flow:

1. The user enters valid login credentials.
2. The system verifies the credentials, account status, assigned role, and branch access.
3. The system creates an authenticated user session.
4. The system opens the dashboard corresponding to the user's role.
5. The user accesses the authorized system functions.

Alternative and Extension Flows:

1. If the credentials are invalid, the system denies access and prompts the user to try again.
2. If the account is inactive, the system informs the user that access is unavailable.
3. The user may manage and update their own profile information.
4. The user may log out, after which the system ends the session and returns to the login page.

Postconditions: A secure user session is created or terminated, and any authorized profile changes are saved.

Use Case 2: Manage Customer and Vehicle Records

Use Case Goal: To maintain accurate customer, vehicle, service, ownership, and branch-related records.

Actor: Front Desk Staff

Preconditions: The user is authenticated and has access to the Customer and Vehicle Records module.

Trigger: The user opens the Customer and Vehicle Records module.

Included Use Case: View Service and Ownership History

Extension Use Cases: Archive Branch Customer Record; Archive Vehicle Record; Restore Vehicle Record; View Cross-Branch Activity

Main Flow:

1. The user searches for an existing customer or creates a new customer record.
2. The system validates and saves the customer information.
3. The user registers or updates the customer's vehicle information.
4. The system validates and saves the vehicle record.
5. The system links the customer and vehicle to the appropriate branch records.
6. The system displays the vehicle's service and ownership history.

Alternative and Extension Flows:

1. If a possible duplicate customer or vehicle is detected, the system displays the existing record for verification.
2. If required information is missing, the system prompts the user to complete the fields.
3. The user may archive a customer's branch record without deleting the centralized customer record.
4. The user may archive an inactive vehicle record.
5. The user may restore an archived vehicle record.
6. The user may view authorized cross-branch customer and vehicle activity.

Postconditions: Customer and vehicle records are created or updated, and their branch, ownership, service history, and archive status are maintained.

Use Case 3: Create Service Operation

Use Case Goal: To record vehicle inspection findings, select required services or items, and create a service operation.

Primary Actor: Front Desk Staff

Supporting Actor: Admin / Owner

Preconditions: The user is authenticated, and the customer and vehicle records exist.

Trigger: The Front Desk Staff initiates a new service operation.

Included Use Case: Calculate Operation Total

Extension Use Cases: Edit Service Operation Items; Archive Pending Service Operation; Request Other-Branch Item; Print Service Operation; Approve / Reject Service Operation

Main Flow:

1. The Front Desk Staff selects the customer and vehicle.
2. The user records the customer's concerns and vehicle inspection findings.
3. The user selects the required services, parts, or inventory items.
4. The system retrieves the corresponding prices and quantities.
5. The system calculates and displays the total service operation amount.
6. The user reviews and saves the service operation.

7. The system records the service operation with a pending status.

Alternative and Extension Flows:

1. The user may edit the service operation items before approval.
2. If an item is unavailable at the current branch, the user may request the item from another branch.
3. The user may print the service operation.
4. The user may archive a pending service operation.
5. The Admin / Owner may approve or reject the pending service operation.
6. If required information is incomplete, the system prevents saving and displays the fields that require correction.

Postconditions: The service operation is saved as pending, approved, rejected, or archived and remains linked to the customer and vehicle.

Use Case 4: Monitor Inventory and Stock Movement

Use Case Goal: To maintain inventory items and monitor stock quantities and movements at each branch.

Actors: Admin / Owner, Front Desk Staff

Preconditions: The user is authenticated and authorized to access inventory functions.

Trigger: The user opens the Inventory and Stock Movement module.

Included Use Case: Add Inventory Item

Extension Use Cases: View Inventory Transactions; Record Initial Stock; Tag Stock Out to Customer or Vehicle; Request Inter-Branch Stock Transfer

Main Flow:

1. The user opens the inventory records for the assigned or selected branch.
2. The system displays the inventory items and current stock quantities.
3. The user enters the information for a new inventory item.
4. The system validates and saves the inventory item.
5. The system displays the updated inventory record.

Alternative and Extension Flows:

1. The user may view the transaction history of an inventory item.
2. The user may record the item's initial stock quantity.
3. The user may record a stock-out transaction and tag it to a specific customer or vehicle.
4. If the current branch has insufficient stock, the user may request an inter-branch stock transfer.

5. For an inter-branch request, the system checks other-branch availability and notifies the affected branches.
6. If duplicate or incomplete item information is entered, the system displays an appropriate warning.

Postconditions: Inventory records and stock movements are saved, and any inter-branch request is recorded and communicated to the affected branches.

Use Case 5: Monitor and Process Inter-Branch Transfers

Use Case Goal: To review, approve, and complete inventory transfers between branches.

Actor: Admin / Owner

Preconditions: The user is authenticated, and an inter-branch item request exists.

Trigger: The Admin / Owner opens the Inter-Branch Transfers module.

Included Use Case: Review Incoming Item Requests

Extension Use Cases: Approve Transfer Request; Complete Transfer Request

Nested Included Use Case: Deduct Donor Stock

Nested Extension Use Case: Receive Stock at Requesting Branch

Main Flow:

1. The system displays incoming item requests.
2. The Admin / Owner selects and reviews a request.
3. The system displays the requested item, quantity, requesting branch, donor branch, and availability.
4. The system records the review status of the request.

Alternative and Extension Flows:

1. The Admin / Owner may approve a transfer request when sufficient donor stock is available.
2. An approved request may be marked as completed when the item is released for transfer.
3. When the transfer is completed, the system deducts the transferred quantity from the donor branch.
4. When the item arrives, the requesting branch may record its receipt, and the system adds the quantity to its stock.
5. If the donor branch has insufficient stock, the system prevents approval and displays an availability warning.

Postconditions: The transfer request status is updated, donor stock is deducted when completed, and requesting-branch stock is updated when receipt is recorded.

Use Case 6: Create Job Order and Update Status

Use Case Goal: To convert an approved service operation into a job order and monitor its progress until completion.

Actor: Front Desk Staff

Preconditions: The user is authenticated, and an approved service operation exists.

Trigger: The user selects an approved service operation for job order creation.

Included Use Case: Generate Service History

Extension Use Cases: Track Task-Level Progress; Notify Customer for Pickup; Archive Waiting Job Order

Nested Included Use Cases: Check Transfer Readiness; Queue Pickup SMS Message

Nested Extension Use Case: Deduct Inventory for Completed Task

Main Flow:

1. The user selects an approved service operation.
2. The system loads the customer, vehicle, service, and item details.
3. The user assigns the technician, schedule, and expected duration.
4. The system creates the job order and its task checklist.
5. The system records the initial job order status.
6. The user updates the job order until the required work is completed.
7. The system generates and saves the vehicle service history.

Alternative and Extension Flows:

1. The user may track the progress of individual tasks.
2. Before updating a task, the system checks whether required inter-branch transfers are ready.
3. When a task using inventory is completed, the system may deduct the required inventory quantity.
4. When the job order is ready for pickup, the user may notify the customer, and the system queues a pickup SMS message.
5. A job order that remains in waiting status may be archived.
6. If the service operation is not approved, the system prevents job order creation.

Postconditions: The job order and task statuses are saved, inventory is updated when applicable, and the completed service is recorded in the vehicle's service history.

Use Case 7: View Service Status Board

Use Case Goal: To provide a centralized summary of current service operations and job order statuses.

Actors: Admin / Owner, Front Desk Staff

Preconditions: The user is authenticated and authorized to view service records.

Trigger: The user opens the Service Status Board.

Included Use Case: View Status Summary

Main Flow:

1. The user opens the Service Status Board.
2. The system retrieves the authorized service operations and job orders.
3. The system groups the records according to their current statuses.
4. The system displays the status summary.
5. The user reviews the current service workload and progress.

Alternative Flow:

1. If no service records are available, the system displays an empty status summary.
2. If the user has branch-limited access, the system displays only records from the assigned branch.

Postconditions: The user views an updated summary of the authorized service operations and job orders.

Use Case 8: Forecasting and Decision Support System

Use Case Goal: To analyze inventory data and generate stock-risk classifications and recommendations.

Actor: Admin / Owner

Preconditions: The user is authenticated, and sufficient inventory and transaction data are available.

Trigger: The Admin / Owner opens the Forecasting and Decision Support System module.

Included Use Cases: Select Weekly, Monthly, or Item Forecast; Apply Branch, Category, Status, Year, and Search Filters; Classify Stock Risk

Nested Included Use Cases: Recommend Reorder Quantity; Recommend Branch Transfer Matches

Main Flow:

1. The Admin / Owner selects a weekly, monthly, or item forecast.
2. The user applies the required branch, category, status, year, and search filters.
3. The system retrieves the matching inventory and transaction data.

4. The system analyzes stock usage and current quantities.
5. The system classifies the stock risk of each applicable item.
6. The system recommends reorder quantities.
7. The system identifies possible branch transfer matches.
8. The system displays the forecast and recommendations.

Alternative Flow:

1. If insufficient historical data are available, the system informs the user that the forecast may be limited.
2. If no records match the filters, the system displays an empty result.

Postconditions: Forecast results, stock-risk classifications, reorder quantities, and branch transfer matches are displayed.

Use Case 9: View Reports and Analytics

Use Case Goal: To generate and review authorized operational, inventory, and archived-record reports.

Actors: Admin / Owner, Front Desk Staff

Preconditions: The user is authenticated and has permission to access reports.

Trigger: The user opens the Reports and Analytics module.

Included Use Case: Select Report Type

Extension Use Cases: View Archived Records Report; View Stock Movement Report; Export CSV

Main Flow:

1. The user opens the Reports and Analytics module.
2. The user selects an available report type.
3. The system retrieves the authorized records.
4. The system processes and displays the report results.
5. The user reviews the generated report and analytics.

Alternative and Extension Flows:

1. The user may view an Archived Records Report.
2. The user may view a Stock Movement Report.
3. The user may export the displayed report in CSV format.
4. If no records match the selected report, the system displays an empty report.
5. If the user lacks permission for a report, the system denies access.

Postconditions: The selected report is displayed and may be exported as a CSV file.

Use Case 10: View Notifications

Use Case Goal: To allow users to review and manage system-generated notifications.

Actors: Admin / Owner, Front Desk Staff

Preconditions: The user is authenticated.

Trigger: The user opens the Notifications module.

Included Use Case: Mark Notification Read or Unread

Extension Use Cases: Receive Low Stock Notice; Receive Pending Operation Notice; Receive Active Job Notice; Receive Transfer Notice

Main Flow:

1. The user opens the Notifications module.
2. The system retrieves the notifications available to the user.
3. The system displays each notification with its type, date, message, and read status.
4. The user selects a notification.
5. The user marks the notification as read or unread.
6. The system saves the updated notification status.

Alternative and Extension Flows:

1. The system may generate a low-stock notice when an item reaches its threshold.
2. The system may generate a pending operation notice.
3. The system may generate an active job notice.
4. The system may generate a transfer notice when an inter-branch request is created or updated.
5. If no notifications are available, the system displays an empty notification list.

Postconditions: The user views the notifications, and any changes to their read or unread status are saved.

Use Case 11: Manage Service Catalog

Use Case Goal: To maintain the services and corresponding information used in service operations.

Actor: Admin / Owner

Preconditions: The user is authenticated and has administrative privileges.

Trigger: The Admin / Owner opens the Service Catalog module.

Included Use Cases: Add Service; Edit Service

Extension Use Case: Archive Service

Main Flow:

1. The Admin / Owner opens the Service Catalog module.
2. The system displays the existing services.
3. The user adds a new service or selects an existing service.
4. The user enters or edits the service information and price.
5. The system validates the information.
6. The system saves and displays the updated service catalog.

Alternative and Extension Flows:

1. The user may archive a service that is no longer offered.
2. If duplicate service information is detected, the system displays a warning.
3. If required information is missing, the system prevents saving and identifies the incomplete fields.

Postconditions: The service catalog is updated, and archived services remain stored for historical reference.

Use Case 12: Manage Users and Roles

Use Case Goal: To maintain user accounts and control their roles and branch access.

Actor: Admin / Owner

Preconditions: The user is authenticated and has administrative privileges.

Trigger: The Admin / Owner opens the Users and Roles module.

Included Use Case: Assign Role and Branch

Extension Use Case: Archive or Reactivate User

Main Flow:

1. The Admin / Owner opens the Users and Roles module.
2. The system displays the registered user accounts.
3. The user creates a new account or selects an existing account.
4. The Admin / Owner enters or updates the account information.
5. The Admin / Owner assigns the appropriate role and branch.
6. The system validates and saves the user account.

Alternative and Extension Flows:

1. The Admin / Owner may archive an active user account.
2. The Admin / Owner may reactivate an archived user account.
3. If a username already exists, the system prompts the user to enter a different username.
4. If the role or branch assignment is missing, the system prevents saving.

Postconditions: The user account, assigned role, branch, and account status are updated.

Use Case 13: Manage Branches and Technicians

Use Case Goal: To maintain branch information and the technician roster of each branch.

Actor: Admin / Owner

Preconditions: The user is authenticated and has administrative privileges.

Trigger: The Admin / Owner opens the Branches and Technicians module.

Included Use Case: Maintain Technician Roster

Extension Use Case: Archive or Reactivate Branch

Main Flow:

1. The Admin / Owner opens the Branches and Technicians module.
2. The system displays the registered branches.
3. The user creates a branch or selects an existing branch.
4. The user enters or updates the branch information.
5. The user adds or updates the technicians assigned to the branch roster.
6. The system validates and saves the branch and technician information.

Alternative and Extension Flows:

1. The Admin / Owner may archive an inactive branch.
2. The Admin / Owner may reactivate an archived branch.
3. If duplicate branch information is detected, the system displays a warning.
4. If required branch or technician information is missing, the system prevents saving.

Postconditions: The branch information and technician roster are updated, and the branch status is maintained.

Use Case 14: Configure System Settings

Use Case Goal: To maintain the organization's branding and contact information displayed by the system.

Actor: Admin / Owner

Preconditions: The user is authenticated and has administrative privileges.

Trigger: The Admin / Owner opens the System Settings module.

Included Use Case: Update Branding and Contact Details

Extension Use Case: Upload Company Logo

Main Flow:

1. The Admin / Owner opens the System Settings module.
2. The system displays the current branding and contact details.
3. The user updates the organization name, address, contact information, or other branding details.
4. The system validates the entered information.
5. The system saves and applies the updated settings.

Alternative and Extension Flows:

1. The Admin / Owner may upload a company logo.
2. If the logo has an unsupported file type or size, the system rejects the upload and displays a warning.
3. If required information is missing, the system prevents saving and identifies the incomplete fields.

Postconditions: The updated branding, contact details, and company logo are saved and applied to the system.